

Mitigating Cyber Threats: Machine Learning and Explainable AI for Phishing Detection

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Abstract

The exponential growth of organizations and users has accelerated the adoption of new technologies, increasing the complexity of online security. Phishing attacks have surged significantly in 2024, with over 932,923 incidents reported in Q3 alone, driven by advanced AI-enabled social engineering tactics. From simple scams to sophisticated schemes exploiting emails, URLs, text messages, and social media platforms, phishing attacks deceive victims into disclosing sensitive information or inadvertently installing malware, often compromising devices as part of more extensive botnet networks. Despite advancements in Cyber-security measures, phishing remains a critical threat, causing substantial financial and reputational damage to businesses. Recently, ML algorithms have demonstrated remarkable efficacy in phishing detection; however, many high-performing models operate as black boxes, raising concerns about transparency, interpretability, and trustworthiness—factors essential in high-stakes applications for ensuring reliability, accountability, and regulatory compliance. This research integrates ML techniques with Explainable Artificial Intelligence (XAI) methodologies to address this issue and enhance model interpretability and transparency in phishing detection. The proposed approach employs Extreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (LightGBM), Random Forest, k-Nearest Neighbors (KNN), Twin Support Vector Machine (Twin SVM), and Convolutional Neural Networks (CNN), evaluated across four publicly available datasets to assess performance and interpretability. The research findings reveal that XGBoost achieved the highest accuracy at 99.65%. The Local Interpretable Model-agnostic Explanations (LIME) method was applied to elucidate the importance of feature and model decision-making processes. This comprehensive approach aims to strengthen Cyber-security resilience against phishing threats while promoting model transparency and regulatory compliance.

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1 Introduction

The exponential growth of organizations and users has been accompanied by the rapid evolution of new technologies and methodologies across various domains[1][2]. Every day, countless websites with login pages are launched, many of which attempt to deceive users into revealing personal information. The growing number of websites makes it increasingly difficult to distinguish legitimate platforms from malicious ones[3][4].

Phishing attacks are escalating steadily, with 932,923 incidents reported globally in the third quarter of 2024 an increase from previous quarters [5]. This rise is largely due to advanced methods such as AI-driven social engineering, significantly amplifying financial and operational risks for organizations worldwide [6]. The continuous rise in daily internet usage has substantially broadened the pool of potential targets for cyber attackers [7]. Phishing attacks often originate through deceptive emails, malicious links, or spoofed URLs that imitate trusted sources [8]. Users who engage with such emails or links are often redirected to fraudulent websites embedded with malicious content. As a result, they may unknowingly disclose sensitive information. To appear legitimate, attackers frequently employ long and complex URLs [9].

Historically, the term “phishing” emerged in the U.S. during the 1990s [10].

Recently, attackers have used fraudulent payment methods to create fake accounts on online platforms, which are then exploited to distribute spam within user communities [11].

Over time, attackers have shifted from instant messaging platforms to email-based phishing campaigns, streamlining their tactics and improving their ability to evade detection [12].

Modern phishing is no longer limited to emails and websites it now includes malicious links shared via text messages, tweets, Facebook, and Instagram posts. These links are designed to capture user credentials like usernames and passwords [4].

Phishing attackers often control the fraudulent websites, capturing any passwords entered by victims

[13].

Phishing victims frequently have their devices infected with malware, which can convert them into botnets for further malicious activities [14]. Phishing remains a critical threat to organizations worldwide. According to Cisco, 92% of companies have experienced at least one compromised business email account due to phishing [15]. The Egress’s 2024 Phishing Threat Trends Report revealed that 93% of companies experienced data leaks due to either employee negligence or stolen credentials. Phishing is now a dominant vector for ransomware distribution. A 2022 study noted that 88% of organizations facing an uptick in email threats also suffered ransomware attacks [16]. Google’s efforts to combat phishing resulted in the blocking of nearly 100 million phishing emails daily in early 2022.

In recent reports, [17] phishing attacks result in significant financial losses and reputational damage for businesses, often leading to customer attrition. Studies indicate that companies can lose up to one percent of their customer base following data breaches predominantly caused by phishing.

Phishing attacks are among the most serious cybersecurity threats, aiming to trick users into disclosing sensitive information through deceptive emails or websites [2]. In response, researchers have proposed ML-based systems to detect phishing URLs with increasing success [18][19]. Advances in networking and communication technologies have transformed online activities such as social networking and banking, thereby increasing exposure to cyber-attacks [20]. While user awareness and caution are essential, phishing scams remain difficult to eliminate entirely.

Although ML algorithms have proven highly effective in identifying phishing attacks, their black-box nature raises concerns about trust, interpretability, and transparency, especially in sensitive domains [21][22]. This study addresses these challenges by designing and evaluating nine feature engineering algorithms to improve phishing detection. In addition, it incorporates Explainable Artificial Intelligence (XAI) to shed light on how the models make decisions.

XAI refers to tools and methods that make ML deci-

sions understandable to humans. The goal is to offer researchers and cybersecurity professionals deeper insights into malicious phishing patterns, aiding in the creation of more robust and trustworthy security systems. The outcomes of this study are expected to enhance phishing defenses by increasing transparency and accuracy in ML-based models.

More specifically, ML and Explainable Artificial Intelligence (XAI) play a pivotal role in modern cybersecurity by enabling advanced threat detection, malware classification, and fraud prevention [23]. ML techniques, including deep learning and anomaly detection algorithms, are capable of processing large-scale data to uncover patterns associated with cyber threats such as phishing and zero-day attacks [24].

Explainable Artificial Intelligence (XAI) enhances the interpretability and transparency of cybersecurity models, enabling security analysts to understand the rationale behind alerts and fostering greater trust in automated defense systems [25]. XAI also plays an essential role in regulatory compliance by explaining AI decisions transparently [26]. Overall, XAI ensures transparency, accountability, and trust in critical sectors such as healthcare, finance, law, and autonomous systems.

The novelty of this research lies in its integration of ML with XAI for enhanced phishing detection. The key contributions include:

1. Identification of essential, interpretable URL features contributing to phishing detection.
2. Evaluation of multiple ML algorithms, with XGBoost achieving a peak accuracy of 99.65%, outperforming other common techniques.
3. Use of Local Interpretable Model-Agnostic Explanations (LIME) to provide detailed, instance-level interpretations of model decisions, fostering transparency and trust.
4. Application of four publicly available datasets to ensure comprehensive evaluation and validate the generalizability of the proposed approach across diverse URL types.

The paper is organized as follows. Section 2 presents the background. Then, the related work is

discussed in section 3. Afterwards, section 4 discusses the materials and methods, and finally, the conclusion is presented in the section 5.

2 Background

2.1 Anatomy of a URL

Uniform Resource Locators (URLs), as shown in Fig. 1, identify websites and consist of several parts: the protocol, top-level domain (TLD), domain name, path, parameter, and subdomain. HTTPS, FTP, and SMTP facilitate communication between web servers and clients. The domain name uniquely identifies a website, while the TLD, like "uk" in "video.google.co.uk," categorizes the domain. Subdomains, such as "video" in "video.google.co.uk," organize website sections. The path directs the server to specific files or directories, and query strings, following a question mark, provide parameters to generate content based on user requests dynamically.

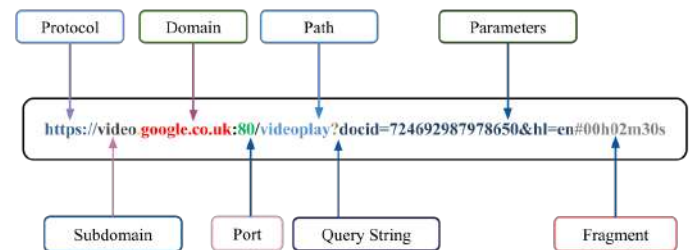


Figure 1. Anatomy of a URL

2.2 Blacklisting phishing URLs

A common approach for detecting phishing URLs is the use of a blacklist, which stores known malicious IP addresses, URLs, and domains associated with phishing activities. When a user accesses a URL, it is checked against this list; if a match is found, the URL is flagged as dangerous. Otherwise, it is considered safe. To maintain effectiveness, blacklist databases are updated frequently often every 12 hours after an initial phishing detection. However, attackers can bypass these systems by making slight alterations to the URL, such as modifying the domain name, file path, or query string. These subtle changes often go undetected by traditional blacklist mechanisms [27].

2.3 Visual Similarity-based Techniques

Many users are deceived by phishing websites due to URLs that closely resemble legitimate ones. It is crucial to verify a site's URL structure and SSL certificate. Modern phishing techniques often replicate legitimate websites by mimicking HTML tags, CSS styles, JavaScript functions, and visual elements such as logos and layouts [28]. A website is considered phishing if it closely resembles a legitimate site. The process of detecting this involves comparing visual elements. Similarity is found by analyzing the extracted information [29] [30] [31].

2.4 Phishing Attacks

Phishing attacks remain a widespread and dangerous threat in cybersecurity, leveraging deceptive emails, messages, or websites to trick individuals into revealing sensitive information such as login credentials, credit card numbers, or personal identification details. Their success largely stems from the exploitation of human psychology, particularly by impersonating trusted entities like financial institutions, social media platforms, or reputable companies. A recent cybersecurity report highlights that phishing was involved in over 36% of data breaches globally in 2023, reflecting its continued dominance in the threat landscape [32]. Attackers often craft convincing messages that create urgency or appeal to curiosity, leading victims to click on malicious links or download malware-laden attachments, thereby compromising their devices and networks [33].

Phishing attacks impact individuals and organizations, leading to financial losses, data breaches, and reputational harm. These attacks can result in unauthorized access to sensitive business information and intellectual property theft, paving the way for further attacks such as ransomware. Research from the Anti-Phishing Working Group (APWG) shows that phishing techniques are becoming more sophisticated, including spear-phishing and clone phishing. Spear-phishing involves personalized messages targeting specific individuals, making it more deceptive, while clone phishing replicates legitimate emails with added malicious content. Organizations are investing in email filtering solutions, employee training on phishing recognition, and

advanced threat detection systems to counter these threats [34][35].

2.5 Explainable Artificial Intelligence

Explainable Artificial Intelligence (XAI) focuses on making the decision-making processes of ML models transparent and understandable to humans [36][25]. As ML systems are increasingly adopted in sensitive and high-stakes domains such as cybersecurity, finance, and healthcare, the demand for interpretability has grown significantly [21]. XAI techniques help build trust by offering insights into how models arrive at specific predictions, which is essential for ensuring reliability, accountability, and fairness. Common XAI strategies include feature importance analysis, surrogate models, and model distillation—all aimed at simplifying the behavior of complex models for human interpretation [37].

Two prominent XAI techniques—Local Interpretable Model Agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP) have shown effectiveness in improving transparency in ML applications.

LIME explains individual model predictions by generating perturbed versions of the input data and fitting a simple interpretable model around the local prediction space [38]. In the context of cybersecurity, LIME has been employed to detect phishing attacks by highlighting key URL features that contribute most to a model's classification decision. This local explanation approach helps security analysts and developers understand and validate why a specific URL was flagged as malicious.

SHAP uses principles from cooperative game theory to assign each feature a Shapley value, representing its contribution to the final prediction [39][40]. SHAP values offer a consistent and theoretically sound measure of feature importance, enabling global and local interpretability across different model types. In phishing detection tasks, SHAP helps uncover which attributes, such as unusual subdomain length or special characters, are driving the model's decisions, aiding in both model validation and feature refinement.

Table 1. Summary of studies on ML, XAI, and phishing detection.

Reference	Authors	Year	Method	Description
[21]	Khan et al.	2024	XAI	Applied SHAP to analyze feature importance in phishing detection models, improving transparency.
[56]	Nwakanma, Ahakonye	2023	XAI for Connected Vehicles	Discusses the application of Explainable AI techniques for intrusion detection and mitigation in intelligent connected vehicles, emphasizing transparency and decision explainability.
[36]	Calzarossa et al.	2023	XAI	Investigated methods like SHAP and LIME to explain AI decisions in phishing detection.
[57]	Zhang, Al Hamadi	2022	XAI in CyberSecurity	Reviews state-of-the-art methods to enhance AI interpretability and trust in cybersecurity applications.
[48]	Chawla et al.	2022	Random Forest Classifier	Developed a phishing detection model with 97.73% accuracy using Random Forest and other ML algorithms.
[47]	Smith et al.	2023	Genetic and Swarm Optimization	Improves phishing detection using genetic algorithms and particle swarm optimization for better classifier performance.
[28]	Zhang et al.	2023	Visual Similarity-based Techniques	Proposed phishing detection method based on structural and visual similarity of HTML tags, CSS, JavaScript, and layout components.
[32]	Verizon	2024	Phishing Attack Analysis	Reported that over 36% of global data breaches in 2023 involved phishing, highlighting its growing impact.
[33]	Smith et al.	2023	Spear-Phishing Analysis	Analyzed spear-phishing and clone phishing as advanced and highly targeted phishing strategies used against individuals and organizations.
[50]	Kumar et al.	2023	Random Forest	Used URL-only features for phishing detection, achieving 97% F1-score, 97% accuracy, and 99% recall.
[42]	K. Lee et al.	2023	NLP-based Detection	Achieved a 99.93% F1-score using XGBoost and natural language processing techniques on phishing email content.
This Work	Usman et al.	2025	Integrating ML with XAI techniques for Phishing Detection	A novel approach to phishing attack detection by integrating ML with XAI techniques, achieving high accuracy and interpretability through comprehensive feature engineering and rigorous evaluation across diverse datasets.

3 Related Work

3.1 Machine Learning In Phishing Attack

The application of ML in phishing detection has attracted growing attention from the research community in recent years [41][42][43][44][45][46]. This surge of interest is driven by the escalating frequency and sophistication of phishing attacks, which demand advanced, adaptive, and scalable detection mechanisms. Owing to their ability to learn from large-scale data and detect complex patterns, ML models provide promising solutions for identifying phishing attempts based on communication structures and network behavior anomalies. Numerous studies have explored and enhanced ML-based detection frameworks to improve accuracy and robustness against evolving threats. In this research, we propose a novel phishing detection method that analyzes hyperlinks embedded in a website's HTML source code to identify phishing traits. We trained and evaluated six ML algorithms using four publicly available datasets and extracted eleven hyperlink-based features. Furthermore, we incorporated LIME to interpret the internal

decision-making of the models, offering transparency and insight into the most influential features, thus increasing the method's trustworthiness.

Smith et al. [47] proposed an improved phishing detection approach by integrating genetic algorithms and particle swarm optimization to enhance classifier performance. Their study revealed that the combination of a multinomial Naive Bayes classifier with a genetic algorithm outperformed other models in terms of detection accuracy. A comprehensive comparative analysis was conducted to benchmark the effectiveness of this hybrid approach against various traditional ML and bio-inspired techniques, demonstrating its superiority in optimizing feature selection and model accuracy. In [48], the authors aimed to identify standard features of phishing websites and create a detection model using feature extraction and pre-processing on seven email datasets, with Random Forest Max Vote Classifier, Decision Tree and Artificial Neural Network models achieving 97.73% accuracy.

Rao and Pais [49] investigated critical website

features, such as URL structure, domain age, SSL certificate status, page length, and the use of JavaScript for effective phishing detection. These features were employed to train various ML models, with Random Forest demonstrating superior performance in accurately classifying phishing websites. Their study confirmed that integrating these lightweight and interpretable features yields high detection accuracy while maintaining computational efficiency. Kumar et al. [50] proposed a lightweight and efficient phishing detection model based solely on URL analysis. Utilizing a Random Forest classifier, their approach achieved an accuracy of 97%, an F1-score of 97%, and a recall of 99%. The study highlighted that this method not only outperformed traditional techniques but also offered faster and more resource-efficient detection by eliminating the need for additional data such as webpage content or user behavior.

In [51], an ML-based approach was introduced for detecting phishing emails using natural language processing techniques. The method achieved a 99.93% F1-score using the XGBoost algorithm, outperforming several state-of-the-art phishing detection models. Notably, the approach relied solely on the content of the email body, eliminating the need for auxiliary features such as headers, IP addresses, or embedded link analysis, thus streamlining detection while maintaining high accuracy. Ali and Malebary[52] proposed a phishing website detection model that incorporates particle swarm optimization (PSO) for feature weighting to enhance classification performance. Their findings demonstrated that applying PSO effectively reduced the feature set while significantly improving detection accuracy and reducing false positives, thereby optimizing the model's overall efficiency.

The Industrial Internet of Things (IoT) is rapidly advancing, connecting digital and physical systems. Its resource-limited devices are vulnerable to security threats. Recent progress in intrusion detection systems (IDS) powered by ML and deep learning [1] provides effective solutions. Arrieta et al.[53] explored IDS advancements, focusing on agent placement, detection techniques, and security challenges while outlining future research directions. In [54], a

hybrid deep learning framework was proposed, combining CNN and BiLSTM models to detect phishing websites using URL-based features. Experimental results showed that the hybrid model achieved higher accuracy and better adaptability to evolving phishing techniques, outperforming traditional ML and earlier deep learning methods.

Recent research highlights the growing use of ML for detecting phishing attacks due to their increasing sophistication and frequency. Studies show that ML models, capable of learning from vast data, offer promising solutions for phishing detection, with approaches such as analyzing website hyperlinks, utilizing genetic algorithms, and particle swarm optimization achieving high accuracy. Notable methods include logistic regression with over 98.4% accuracy, Random Forest classifiers with high F1 scores, and PSO-based feature weighting, significantly improving detection rates while using fewer features.

3.2 XAI in Phishing Attack Detection

The use of XAI is a highly active area of research within the CyberSecurity community, particularly in the context of phishing attack detection[55][56][57][58][59]. Researchers are increasingly focusing on XAI to enhance the transparency and interpretability of ML models used in CyberSecurity applications. The complexity and opacity of traditional black-box models pose significant challenges for security professionals who need to understand and trust the decisions made by these systems. XAI techniques address these challenges by providing insights into how models arrive at their predictions, thereby improving the reliability and accountability of phishing detection systems. Table 1 summarises recent research on ML and XAI in the literature.

In [55] introduced a method for detecting malicious URLs using statistical features from n-grams extracted from hexadecimal URL representations. By employing ML and deep learning models, along with XAI for feature evaluation, the method reduced the number of features and training time compared to previous approaches that rely on numerous high-level URL features. Fan et al. [58] proposed an ML approach using explainable artificial intelligence to analyze how

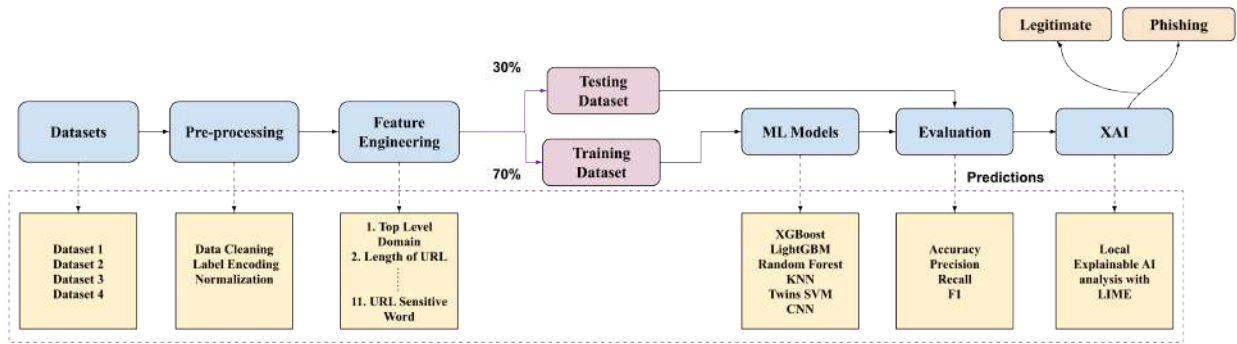


Figure 2. Proposed method for phishing detection and interpretation utilizing ML and XAI.

human and demographic factors influence phishing susceptibility. The model achieved 78% accuracy, 71% recall, and 57% precision and highlighted the impact of psychological characteristics and online security habits on phishing risk, offering personalized recommendations for mitigation based on individual circumstances.

Aslam et al. [59] proposed an XAI model for detecting malicious domains using a dataset of 45,000 samples. Among various models tested, Extreme XGBoost achieved the highest performance, with an accuracy of 98.56% and an AUC of 0.9991 before feature selection, and an accuracy of 98.18% and an AUC of 0.999 after feature selection. Using Shapley additive explanations and LIME to understand the model made it easier to explain XGB's predictions, doing better than earlier studies.

Incorporating XAI into CyberSecurity not only aids in detecting phishing attacks more effectively but also helps understand the underlying mechanisms of these threats. For instance, XAI methods like LIME and SHAP can elucidate which features most indicate phishing attempts, such as suspicious URLs or email metadata. This interpretability is crucial for CyberSecurity experts to develop robust defence strategies and for regulatory compliance, where understanding the rationale behind automated decisions is necessary. As a result, XAI is becoming an essential component of modern CyberSecurity research, driving advancements in detecting and mitigating cyber threats.

Table 2. Summary of datasets utilized in this study.

S#	Dataset	Year	Phishing	Non-Phishing
1	ISCX-URLs	2016	9,965	35,378
2	P.L-URLs	2024	700	300
3	PhishGurad URLs	2024	25,100	35,017
4	Suspicious-URLs	2023	25,000	25,070

4 Materials and Methods

The proposed method for detecting phishing attacks and explaining crucial features in the internal decision-making processes is illustrated in Fig. 2. This study elucidated the significance of features for internal decision-making processes by applying LIME. The presented approach encompasses three significant steps: first, feature engineering from raw URLs; second, training ML models; and third, employing LIME techniques to interpret the models and visualize their internal decision-making processes. All ML models and their corresponding datasets are publicly accessible via the following link: ¹

4.1 Data Collection and Pre-Processing

To demonstrate the effectiveness of the proposed approach, Table 2 outlines the datasets employed in this study for phishing detection, detailing their year of publication, as well as the number of phishing and non-phishing URLs. The ISCX-URLs dataset, available at <https://www.unb.ca/cic/datasets/url-2016.html>, contains 9,965 phishing URLs and 35,378 non-phishing URLs. It is a comprehensive resource encompassing various

¹https://drive.google.com/drive/u/0/folders/1GGKcjk_ZErpA9LS2Y1uAYx5BbT6Sr2-R

Table 3. Overview of engineered features extracted from the datasets along with their descriptions.

Sr.	Feature	Description
1	No. of Top-Level Domains	Counts the number of TLDs in the URL, which can reveal compound or redirected domains.
2	Length of URL	Measures the total character count of the URL, which may suggest its complexity or intent.
3	No. of Digits in Query	Counts digits in the URL's query string, potentially indicating tracking or complex data manipulation.
4	No. of Dots in URL	Counts periods in the URL, which can reflect the structure and potential for obfuscation.
5	No. of Delimiters in Domain	Counts delimiters in the domain, which can indicate attempts to manipulate the domain's appearance.
6	No. of Delimiters in Path	Counts delimiters in the URL path, suggesting the depth and complexity of the URL structure.
7	Length of Longest Token in Path	Measures the longest segment in the URL path, which may house significant or suspicious elements.
8	Domain Length	Measures the character count of the domain name, which can be indicative of the site's legitimacy.
9	Entropy of Domain	Assesses the randomness of the domain name, with higher entropy potentially indicating a malicious domain.
10	Domain Length Density	Ratio of the domain length to the total URL length, which can highlight the proportion of the domain in the URL.
11	URL Sensitive Word	Detects the presence of sensitive keywords in the URL that may indicate phishing or fraudulent activity.

URLs, including spam records, phishing links, malware, and benign URLs, making it highly suitable for training and evaluation. The second dataset PL-URLs, is accessible at <https://data.mendeley.com/datasets/j43jtv3zzc/1>, comprises 700 phishing and 300 non-phishing URLs. Although smaller, its balanced composition makes it an important resource for assessing model performance under controlled conditions with minimal bias.

The third dataset PhishGuard URLs, available at <https://data.mendeley.com/datasets/vfszjb9b36/1>, includes 25,100 phishing URLs and 35,017 non-phishing URLs. Its extensive size and diverse content allow for a comprehensive model generalization and robustness evaluation. The final dataset Suspicious-URLs, published in Frontiers in Computer Science with DOI [10.3389/fcomp.2024.1308634](https://doi.org/10.3389/fcomp.2024.1308634), contains 25,000 phishing URLs and 25,070 non-phishing URLs. Its balanced distribution provides a valuable resource for analyzing model performance against contemporary phishing threats.

These datasets collectively provide a diverse and robust foundation for evaluating the proposed phishing detection methodology, enabling the assessment of model effectiveness across various data distributions, URL types, and timeframes. Poor data pre-processing can introduce biases and inconsistencies, potentially leading to flawed assumptions and

misleading outcomes. Adequate data preparation involves critical tasks such as cleaning, integration, dimensionality reduction, and discretization. In the context of phishing detection, the primary goal of data pre-processing is to reduce false positives and false negatives [60][61].

During pre-processing, data cleaning and feature scaling operations were performed. Feature scaling ensures that all features are normalized to a standard range, which prevents any single variable from disproportionately influencing the model. This process standardizes the data distribution across features, improving the model's performance and consistency.

4.2 Feature Engineering

Feature engineering is an important step in the ML process. It means creating or changing new features to help the model perform better. The goal is to turn raw data into a form the model can understand more easily, which helps it make more accurate predictions. This can include scaling numbers, converting categories into numbers, combining features, or adding new features based on expert knowledge. In short, feature engineering helps give the model the best possible information to learn from [62][63].

4.3 Feature Engineering Algorithms

In this research, to better analyze the structure and behavior of URLs, 11 features were selected and extracted from the datasets. These features focus on the

basic structure, such as the number of dots, delimiters, and URL length, and more advanced indicators, like domain entropy and sensitive words. Table 3 provides an overview of these features and brief descriptions, highlighting how each can help identify potentially malicious or phishing URLs.

Algorithm 1. Count the Number of Tokens in the URL

Require: URL

Ensure: the Number of token counts in the URL

```

1: function tokenCount(URL)
2:   count  $\leftarrow$  0
3:   index  $\leftarrow$  0
4:   Tokenized_list  $\leftarrow$  tokenize(URL)
5:   for all token in Tokenized_list do
6:     count  $\leftarrow$  count + 1
7:     D[index]  $\leftarrow$  count
8:   end for
9: end function

```

Feature engineering often involves breaking raw data into useful components. For instance, in phishing detection, a URL like `http://abc.edu/last/` can be tokenized into parts such as ['http', 'abc', 'edu', 'last']. These tokens help identify patterns that distinguish malicious URLs from legitimate ones. In the proposed approach, the first step was to break the URL into smaller parts. This process is called tokenization. Breaking the URL like this helped in finding useful patterns to check if a URL is safe or dangerous. These parts (tokens) are then used to make a feature vector for the ML model. This helps the model detect phishing websites better. For example, in algorithm 1, the tokenize function creates a list called *tokenized_list* that stores all the parts. The total number of parts is saved in *D*, and the feature vector keeps track of each part and its position.

The computational procedures for the eleven selected features are detailed in the following sections to provide a comprehensive overview of the feature engineering process.

4.3.1 Number of top-level-domain

In recent observations, attackers have utilized multiple top-level domains (TLDs) within a single domain name to obfuscate their true intentions. This tool systematically tracks the number of TLDs present in a given URL. The extraction of this feature from URLs is outlined in Algorithm 2. In our methodology, we monitored a comprehensive list of 920 TLDs, denoted as $P = [P_1, P_2, P_3, \dots, P_y]$.

While a standard URL u is expected to contain a single TLD, represented as $d = [d_1, d_2, d_3, \dots, d_n]$, there are cases where a URL may incorporate multiple TLDs. For each segment d_k in the domain, where d_k is any potential TLD, it is checked against the list P . The algorithm proceeds by extracting the TLD from the URL and determining the frequency of its occurrence. The valid TLDs identified at each step are stored in the list D . The presence and count of each TLD are then recorded in dataset D , providing a measure of TLD multiplicity in the analyzed URLs.

Algorithm 2. Find Number of Top-Level Domains in URL

Require: URL

Ensure: Number of top-level domains in URL

```

1: function find_domain(URL)
2:    $P \leftarrow \{P_1, P_2, \dots, P_n\}$   $\triangleright$  List of known TLDs
3:   count  $\leftarrow$  0
4:   index  $\leftarrow$  0
5:    $d \leftarrow$  find_top_level_domain(URL)
6:   for all domain in  $d$  do
7:     if domain  $\in P$  then
8:       count  $\leftarrow$  count + 1
9:       D[index]  $\leftarrow$  count
10:    end if
11:  end for
12: end function

```

It is observed that attackers often use multiple top-level domains (TLDs) in a single URL to hide their intent. An algorithm extracts and counts TLDs from URLs by checking each segment against 920 valid TLDs. The identified TLDs are recorded, helping detect suspicious patterns in malicious URLs.

4.3.2 Length of URL

Phishing URLs tend to be significantly longer than legitimate ones, as attackers often incorporate multiple domain names and extended links to obfuscate the URL's true nature and intentions. This deliberate manipulation increases the overall length of the URL, making it more challenging for users and security systems to detect malicious activity. Algorithm 3 calculates the total length of the given URL by systematically counting every character within it.

Algorithm 3. Calculate Length of URL

Require: URL

Ensure: Length of URL

```

1: function length(URL)
2:    $length \leftarrow 0$ 
3:    $index \leftarrow 0$ 
4:   for all character in URL do
5:      $length \leftarrow length + 1$ 
6:      $D[index] \leftarrow length$ 
7:   end for
8: end function
```

Phishing URLs are often longer than legitimate ones. The algorithm counts each character in the URL to determine its total length, helping to identify unusually long URLs that may indicate malicious activity.

4.3.3 Number of digits in Query

The algorithm tokenizes the text and appends a question mark before searching for numeric values within the query string of a URL. The notation $Q = [q_1, q_2, q_3, \dots, q_n]$ indicates that Algorithm 4 may encounter multiple queries associated with a single URL u . The set $X = [x_1, x_2, x_3, \dots, x_n]$ represents the sum of all numeric values found within each query in Q .

The function responsible for counting digits within these queries calculates the total number of digits across all elements of Q . For a given URL u , let Y denote the total number of characters. The sum x_k for each query k is determined by summing the digits found in that specific query. This relationship can be

mathematically defined as follows:

$$Y = \sum_{k=1}^n, \quad (1)$$

where n is an arbitrary number.

Algorithm 4. Number of Digits in Query String

Require: URL

Ensure: the Number of digits in a query string

```

1: function digits_in_query_string(URL)
2:    $y \leftarrow 0$ 
3:    $index \leftarrow 0$ 
4:    $Q \leftarrow \text{query\_string}(\text{URL})$ 
5:   if  $Q$  is not empty then
6:     for all query in  $Q$  do
7:        $X_k \leftarrow \text{digits\_in\_query\_string}(\text{query})$ 
8:        $y \leftarrow y + X_k$ 
9:     end for
10:     $D[index] \leftarrow y$ 
11:   else
12:     $D[index] \leftarrow -1$ 
13:   end if
14: end function
```

4.3.4 Number of dots in URL

The number of dots in a URL can be an important indicator of its authenticity. It is observed that phishing websites often use more dots to create the illusion of legitimacy, deceiving users into believing they are visiting a genuine site. For example, a legitimate URL like "www.abc.com" could be manipulated to appear as "www.ab.c.com" misleading users. Recognizing and being cautious of URLs with unusual dot patterns is key to avoiding phishing scams.

4.3.5 Number of delimiters in domain

Attackers sometimes employ special characters, such as '+' or '-', to make a phishing URL appear legitimate. For example, while the legitimate URL might be "http://www.abc.com/" a phishing variant could be crafted as 'www.ab+c.com' to deceive users into thinking it is authentic. Recognizing these patterns and understanding the role of such delimiters is crucial in identifying and avoiding phishing attempts.

Algorithm 5. Length of the Longest Token in the Path**Require:** URL**Ensure:** Length of the longest token in the URL path

```

1: function longest_path(URL)
2:    $index \leftarrow 0$ 
3:    $G \leftarrow \text{find\_paths}(\text{URL})$ 
4:   if  $G$  is not empty then
5:     Initialize empty list  $F$ 
6:     for all path in  $G$  do
7:        $f_k \leftarrow \text{find\_length}(\text{path})$ 
8:       Append  $f_k$  to  $F$ 
9:     end for
10:     $H \leftarrow \max(F)$ 
11:     $D[m_7][index] \leftarrow H$ 
12:  else
13:     $D[m_7][index] \leftarrow -1$ 
14:  end if
15: end function

```

In this method, six characters—specifically ‘:’, ‘/’, ‘.’, ‘|’, and ‘+’—are used as separators to identify and parse different URL segments. The find-available-domain method determines the number of domains within a phishing URL by comparing these characters to the set V . The delimiter-count method increases the count D each time a match is found between the characters in the URL and the set V .

4.3.6 Number of delimiters in path

When comparing URLs, it was observed that phishing paths often contain twice as many spaces between words compared to legitimate URLs. It is observed that Phishing paths frequently include potentially harmful files, such as ‘.png’ or ‘.js’, following delimiters, which can facilitate malicious activities. In the proposed algorithm 6, the “find-available-path” method identifies the total number of path segments in a URL, while the marker count function tracks the locations where these breaks occur within the path. This analysis is critical for detecting the subtle differences in URL structure that indicate phishing attempts.

4.3.7 Length of the longest token in path

In general, Attackers often conceal malicious content within long and complex URL codes, making it significantly more time-consuming to access phishing URLs—approximately three times longer than accessing legitimate URLs. The find-paths function in Algorithm 6 identifies multiple paths within the URL list G , while the find-length function calculates the length of each path, storing the longest path length in H . For URLs that do not contain a valid path, a value of ‘-1’ is recorded in dataset D . This approach helps detect URLs with suspiciously long and convoluted structures, often indicative of phishing attempts.

4.3.8 Domain length

Attackers often use elongated domains to create the appearance of a legitimate URL, while genuine URLs typically feature shorter, more concise domains. This feature compares the domain lengths of authentic and fake URLs. By determining the domain length of each URL and storing these values in a collection, we can analyze the differences between complete and fraudulent URLs. This comparison helps identify potentially deceptive domains artificially lengthened to mislead users.

For instance, attackers often extend domain names to disguise phishing URLs as legitimate. This method compares the length of domains in genuine and suspicious URLs, helping to identify fraudulent ones. The algorithm counts delimiters in the domain and path to detect unusual patterns.

4.3.9 Entropy Domain

Entropy Domain measures the unpredictability of domain names within a URL by calculating the entropy of their frequency distribution. Entropy is computed using the formula:

$$H = - \sum p_i \log_2(p_i), \quad (2)$$

Where p_i represents the probability of each unique domain. The algorithm functions by counting the occurrences of each domain, computing the likelihood for each domain, and subsequently determining the entropy based on these probabilities. Algorithm 7 il-

Algorithm 6. Number of Delimiters in Path and Domain**Require:** URL**Ensure:** Number of delimiters in path and domain

```

1: function DelimiterCount(input)
2:    $V \leftarrow \{'@', '/', '=', '\}$ 
3:    $count \leftarrow 0$ 
4:   for all character in input do
5:     if character  $\in V$  then
6:        $count \leftarrow count + 1$ 
7:     end if
8:   end for
9:   return  $count$ 
10: end function
11: function NumberOfDelimiter(URL)
12:    $C_1 \leftarrow 0, C_2 \leftarrow 0$ 
13:    $pathIndex \leftarrow 0, domainIndex \leftarrow 0$ 
14:    $G \leftarrow \text{findAvailablePath}(\text{URL})$ 
15:    $R \leftarrow \text{findAvailableDomain}(\text{URL})$ 
16:   if  $G$  is not empty then
17:     for all path in  $G$  do
18:        $count \leftarrow \text{DelimiterCount}(\text{path})$ 
19:        $C_1 \leftarrow C_1 + count$ 
20:        $D[m_6][pathIndex] \leftarrow C_1$ 
21:     end for
22:   else
23:      $D[m_6][pathIndex] \leftarrow -1$ 
24:   end if
25:   if  $R$  is not empty then
26:     for all domain in  $R$  do
27:        $count \leftarrow \text{DelimiterCount}(\text{domain})$ 
28:        $C_2 \leftarrow C_2 + count$ 
29:        $D[m_5][domainIndex] \leftarrow C_2$ 
30:     end for
31:   else
32:      $D[m_5][domainIndex] \leftarrow -1$ 
33:   end if
34: end function

```

illustrates the procedure for calculating the entropy of domains.

Algorithm 7. Entropy of Domains**Require:** URL**Ensure:** Entropy of domain names

```

1: function Calculate_Entropy(URL)
2:    $domain\_list \leftarrow \text{extract\_domains}(\text{URL})$ 
3:   if  $domain\_list$  is not empty then
4:     Initialize empty dictionary  $frequency$ 
5:     for all domain in  $domain\_list$  do
6:       Increment  $frequency[\text{domain}]$  by 1
7:     end for
8:      $entropy \leftarrow 0$ 
9:      $total\_domains \leftarrow \text{length of } domain\_list$ 
10:    for all  $freq$  in  $frequency.values()$  do
11:       $prob \leftarrow freq/total\_domains$ 
12:       $entropy \leftarrow entropy - prob \times \log_2(prob)$ 
13:    end for
14:    return  $entropy$ 
15:  else
16:    return -1
17:  end if
18: end function

```

Entropy evaluates the randomness of domain names in a URL by analyzing their frequency distribution. For example, domains are extracted from the URL, their occurrences are counted, and probabilities are calculated. Based on these probabilities, the entropy is determined, with higher values indicating more unpredictability. This measure helps identify patterns that are useful for detecting malicious URLs.

4.3.10 Domain Length Density

Domain Length Density is a feature that measures the length of domain names in a URL and evaluates how this might indicate suspicious activity. A high-density or unusually long domain may suggest phishing. The algorithm calculates the length of each domain within the URL and determines the density by considering the total length of the domain names relative to the number of domains. The algorithm 8 illustrates the procedure for calculating the domain-length density.

Algorithm 8. Domain Length Density

Require: URL
Ensure: Average domain length

```

1: function Calculate_Domain_Length_Density(URL)
2:   domain_list  $\leftarrow$  extract_domains(URL)
3:   total_length  $\leftarrow$  0
4:   domain_count  $\leftarrow$  length of domain_list
5:   if domain_count > 0 then
6:     for all domain in domain_list do
7:       total_length  $\leftarrow$  total_length + length of domain
8:     end for
9:     return total_length / domain_count
10:  else
11:    return -1
12:  end if
13: end function

```

Domain Length Density evaluates the average length of domains in a URL. Unusually long domains may indicate phishing. The algorithm sums the lengths of all domains and divides them by the number of domains to calculate the average. If no domains are found, it returns -1.

4.3.11 URL sensitive Word

URL Sensitive Word is a feature that identifies the presence of sensitive or suspicious words within a URL. Such words often include terms related to financial services, personal data, or other high-risk areas. Detecting these terms can aid in flagging potentially malicious URLs. The algorithm 9 demonstrates the procedure for detecting and utilizing sensitive words in URL analysis. The "URL Sensitive Word" feature counts risky words like "bank" or "password" in a URL to detect suspicious links.

4.4 Performance metrics

The ML classification models were employed to categorize URL instances as "phishing" or "non-phishing". The label "phishing" denotes malicious URLs that pose potential harm to users, whereas "non-phishing" refers to URLs considered safe for browsing. A confusion matrix was computed to assess the classification models'

Algorithm 9. Sensitive Words in URL

Require: URL
Ensure: Number of sensitive words found

```

1: function Detect_Sensitive_Words(URL)
2:   sensitive_words  $\leftarrow$  {list of sensitive words}
3:   count  $\leftarrow$  0
4:   token_list  $\leftarrow$  tokenize(URL)
5:   for all word in token_list do
6:     if word  $\in$  sensitive_words then
7:       count  $\leftarrow$  count + 1
8:     end if
9:   end for
10:  return count
11: end function

```

performance, as illustrated in Table 4. A confusion matrix is used to check how well a classification model works. It has four parts: true positives (TP), false positives (FP), false negatives (FN), and true negatives (TN). True positives are when the model correctly predicts the positive class. False positives are when it wrongly predicts a negative case as positive. False negatives happen when the model misses a positive case, and true negatives are when it correctly predicts the negative class.

Table 4. Confusion matrix.

		Predicted Labels	
		Non-Phishing	Phishing
Actual Labels	Non-Phishing	TP	TN
	Phishing	FP	FN

Several key parameters were selected to evaluate and compare the precision of the classification model. These include the number of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). Metrics such as Accuracy (ACC), Precision (P), Recall (R), Sensitivity (TPR), Specificity (TNR), and the Area Under the ROC Curve (AUC) were used as designated quantitative measures. The total number of instances is denoted as N , and these metrics are computed accordingly to provide a comprehensive assessment of the model's performance. The value N can be calculated as:

$$N = (TP) + (TN) + (FP) + (FN) \quad (3)$$

Sensitivity (TPR): also known as Recall (R), refers to the proportion of true positives correctly identified out of all actual positives. It indicates how well the model identifies positive cases.

$$TPR = \frac{(TP)}{(TP) + (FN)} \quad (4)$$

Specificity (TNR): Specificity is a metric used to evaluate the performance of a classification model, particularly in binary classification tasks. It measures the proportion of true negatives correctly identified by the model out of all negatives. In other words, specificity tells us how well the model avoids false positives.

$$TNR = \frac{(TN)}{(TN) + (FP)} \quad (5)$$

False Positive Rate (FPR): also known as the fall-out, measures the proportion of actual negatives that are incorrectly identified as positives and is defined as:

$$FPR = \frac{FP}{FP + TN} \quad (6)$$

False Negative Rate (FNR): also known as the miss rate, measures the proportion of actual positives that are incorrectly identified as negatives and is defined as:

$$FNR = \frac{FN}{FN + TP} \quad (7)$$

Precision (P): Precision tells you how accurate the model's positive predictions are. It is the proportion of true positives among all positive predictions.

$$P = \frac{(TP)}{(TP) + (FP)} \quad (8)$$

Area Under the ROC Curve (AUC): The ROC curve graphically displays the predictive model's performance by plotting the actual positive rate against the false positive rate for various threshold settings. AUC represents the likelihood that the model will rank a randomly chosen positive instance higher than a randomly chosen negative one. The perfect prediction value for AUC is 1.0, and 0.5 corresponds to the diagonal on the ROC curve.

F-Measure (F1 Score): The harmonic mean of Precision and Recall. It balances the two metrics, especially when there is an uneven class distribution.

$$F1 \text{ Score} = \frac{2 \times (P) \times (R)}{(P) + (R)} \quad (9)$$

Accuracy (ACC): The number of correct predictions a classification algorithm makes is called its accuracy. To find it, divide the number of correct predictions by the total number of data points.

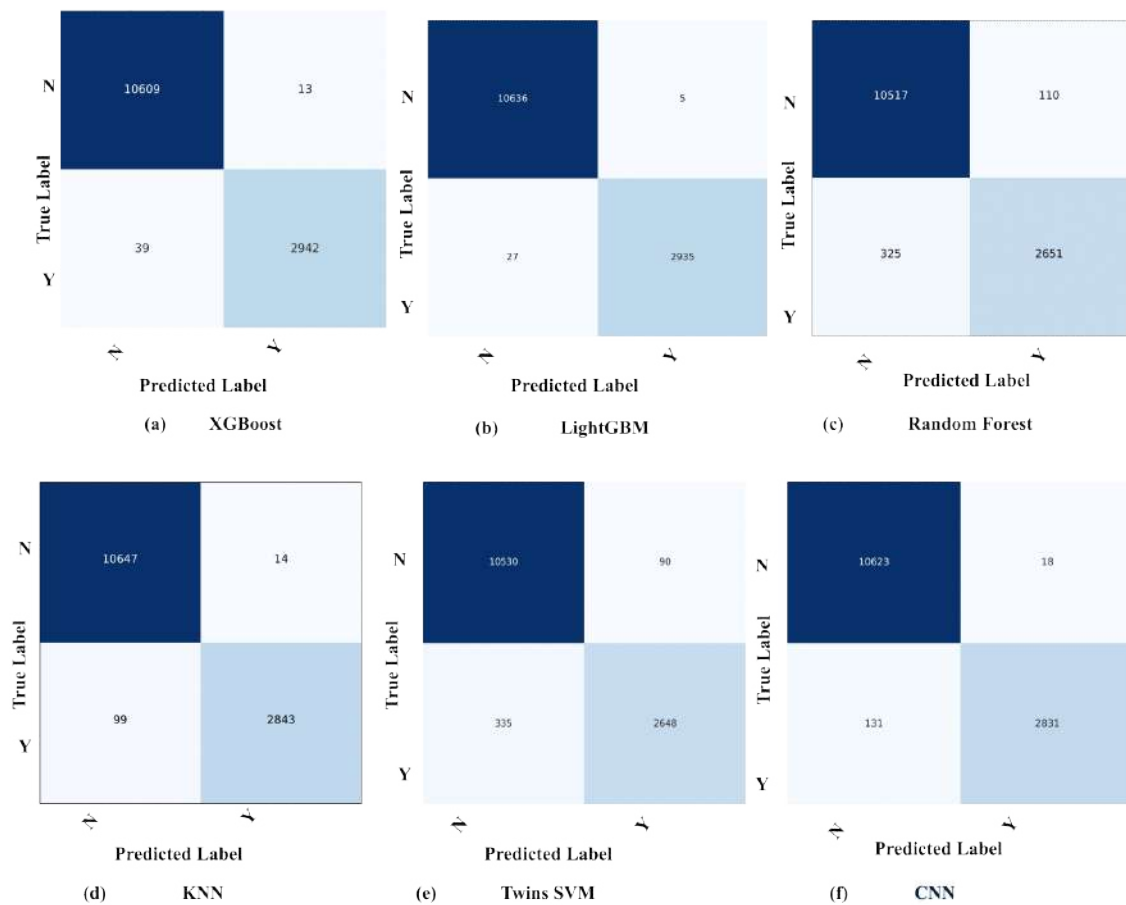
$$\text{Accuracy} = \frac{(TP) + (TN)}{(TP) + (TN) + (FP) + (FN)} \quad (10)$$

These parts impact important metrics. Accuracy shows how often the model is right overall. Precision measures how many of the positive predictions are correct. Recall shows how many actual positive cases the model identifies. The F1-score balances precision and recall, especially with imbalanced data. Specificity measures how well the model avoids false alarms. These insights help improve the model's performance.

4.5 Results

The results presented in Table 5 and Table 6 empirically demonstrate that the proposed approach achieved robust performance across all datasets used in the experiments. After evaluating the performance of the ML models, it is crucial to interpret and analyze the results to gain a comprehensive understanding of the models' behaviour. This entails identifying the key features that substantially influence the predictions, exploring the relationships between these features and the target variable, and detecting significant patterns or trends within the dataset. In this study, six ML models were evaluated, and the model-agnostic interpretability technique, LIME, was employed to support this analysis and provide valuable insights into the models' decision-making processes. Furthermore, the integration of LIME provided visual insights into feature importance, thereby enhancing interpretability and facilitating improved decision-making.

Table 5 focused on critical detection metrics, including TPR, TNR, FPR, and FNR. Among the algorithms evaluated, XGBoost demonstrated consistent superiority across all datasets, achieving high TPR and

**Figure 3.** Confusion Matrix Analysis: ISCX-URLs Dataset (2016)**Table 5.** Performance comparison of (TPR), (TNR), (FPR), and (FNR).

ML Algorithms	ISCX-URLs				PL-URLs				PhishGuard URLs				Suspicious-URLs			
	TPR	TNR	FPR	FNR	TPR	TNR	FPR	FNR	TPR	TNR	FPR	FNR	TPR	TNR	FPR	FNR
XGBoost	99.56	99.63	0.37	0.44	98.55	100.00	0.00	1.45	96.47	94.98	5.02	3.53	99.92	99.26	0.74	0.08
LightGBM	99.83	99.75	0.25	0.17	100.00	98.21	1.79	0.00	96.49	95.08	4.92	3.51	99.93	99.49	0.51	0.07
Random Forest	96.02	97.00	3.00	3.98	98.54	98.44	1.56	1.46	95.45	94.54	5.46	4.55	99.76	99.16	0.84	0.24
KNN	99.51	99.08	0.92	0.49	98.51	92.75	7.25	1.49	95.79	94.26	5.74	4.21	97.41	94.04	5.96	2.59
Twins SVM	96.71	96.92	3.08	3.29	98.08	100.00	0.00	1.92	78.67	88.37	11.63	21.33	97.80	91.68	8.32	2.20
CNN	99.37	98.78	1.22	0.63	99.07	94.64	5.36	0.93	84.1	89.49	10.51	15.9	89.91	76.62	23.38	10.09

TNR with minimal FPR and FNR. For instance, in the Suspicious-URLs dataset, XGBoost achieved a TPR of 99.92% and a TNR of 99.26%, reflecting its reliability in identifying phishing URLs while minimizing errors. In the PL-URLs dataset, XGBoost recorded a perfect TNR and no false positives, further underscoring its precision. LightGBM closely followed XGBoost in

performance, achieving competitive results in most datasets. In the PL-URLs dataset, LightGBM recorded a TPR of 100.00% and a TNR of 98.21%, demonstrating its strong detection capability. Random Forest also performed well but showed a tendency toward higher FPR in certain datasets, such as an FPR of 3.00% in the ISCX-URLs dataset, slightly impacting its overall relia-

Table 6. Performance comparison of Area Under the Curve (AUC), F-Measure, and Accuracy.

ML Algorithm	ISCX-URLs			PL-URLs			PhishGuard URLs			Suspicious-URLs		
	AUC	F-Measure	Accuracy	AUC	F-Measure	Accuracy	AUC	F-Measure	Accuracy	AUC	F-Measure	Accuracy
XGBoost	99.60	99.12	99.65	99.28	99.27	100.00	95.72	94.62	95.58	99.59	99.58	99.59
LightGBM	99.79	99.47	99.77	99.11	99.77	99.57	95.79	94.66	95.65	99.71	99.7	99.71
Random Forest	96.51	92.42	99.59	98.49	99.02	99.25	94.99	93.82	94.91	99.46	99.45	99.45
KNN	99.29	98.05	99.21	95.63	98.03	98.13	95.02	93.75	94.87	95.72	95.56	95.65
Twins SVM	96.81	92.57	96.86	99.04	99.03	99.63	83.52	81.60	83.99	94.74	94.31	94.52
CNN	99.08	97.44	98.9	96.86	98.79	97.67	86.79	84.73	87.21	83.27	79.75	81.89

bility. K-Nearest Neighbor exhibited variability in its results, with strong TPR values but inconsistent TNR. For example, in the PL-URLs dataset, KNN recorded an FPR of 7.25%, indicating occasional false detections. Twins SVM and CNN struggled with certain datasets, reflecting their limitations in handling complex URL patterns. Twins SVM, for instance, recorded the lowest TPR of 78.67% and the highest FNR of 21.33% in the PhishGuard URLs dataset, while CNN experienced higher FPR and FNR in datasets like Suspicious-URLs, where its FPR reached 23.38%.

Table 6 provides a higher-level comparison of model performance using metrics such as AUC, F-Measure, and Accuracy. XGBoost again emerged as the most robust algorithm, achieving the highest scores across all datasets. In the ISCX-URLs dataset, it recorded an AUC of 99.60%, an F-Measure of 99.12%, and an Accuracy of 99.65%. Even in challenging datasets such as PhishGuard URLs, XGBoost maintained strong results with an AUC of 95.72% and an F-Measure of 94.62%. LightGBM closely rivalled XGBoost in performance, achieving an AUC of 99.71% and an F-Measure of 99.70% in the Suspicious-URLs dataset. While reliable, Random Forest recorded slightly lower F-Measure and AUC values than XGBoost and LightGBM. KNN displayed strong potential in specific datasets, such as an AUC of 99.29% in the ISCX-URLs dataset, but its performance was less consistent across other datasets.

The experimental results demonstrated that Twins SVM and CNN demonstrated reduced effectiveness, with lower AUC, F-Measure, and Accuracy values in several cases. For instance, Twins SVM recorded an F-Measure of 81.60% and an Accuracy of 83.99% in

the PhishGuard URLs dataset, highlighting its limitations in complex detection scenarios. Similarly, CNN struggled with variability in performance, achieving an F-Measure of 79.75% and an Accuracy of 81.89% in the Suspicious-URLs dataset. Overall, XGBoost consistently delivered the best results across all datasets, establishing itself as the most reliable and effective algorithm for phishing URL detection. LightGBM provided a strong alternative, while Random Forest and KNN offered competitive but more variable performance. Twins SVM and CNN exhibited significant limitations, underscoring the importance of selecting robust algorithms for cybersecurity applications.

The confusion matrix in Fig. 3 provides a visual representation of the classification results for each algorithm, illustrating the distribution of True Positives, False Positives, True Negatives, and False Negatives. This matrix is crucial for understanding the exact nature of each model's errors and successes, offering insights into their overall reliability. ROC curves Fig.4 illustrate the trade-off between TPR and FPR across all datasets, with AUC as the key performance metric. XGBoost consistently achieved the highest AUC values, demonstrating superior accuracy and reliability in distinguishing phishing from non-phishing URLs. Although LightGBM and Random Forest performed well in some datasets, models like KNN, Twins SVM, and CNN showed lower AUCs due to higher FPRs. Overall, XGBoost proved to be the most robust and effective model across all datasets, as reflected in Fig.5, which highlights its superior ability to balance precision and recall while maintaining low error rates.

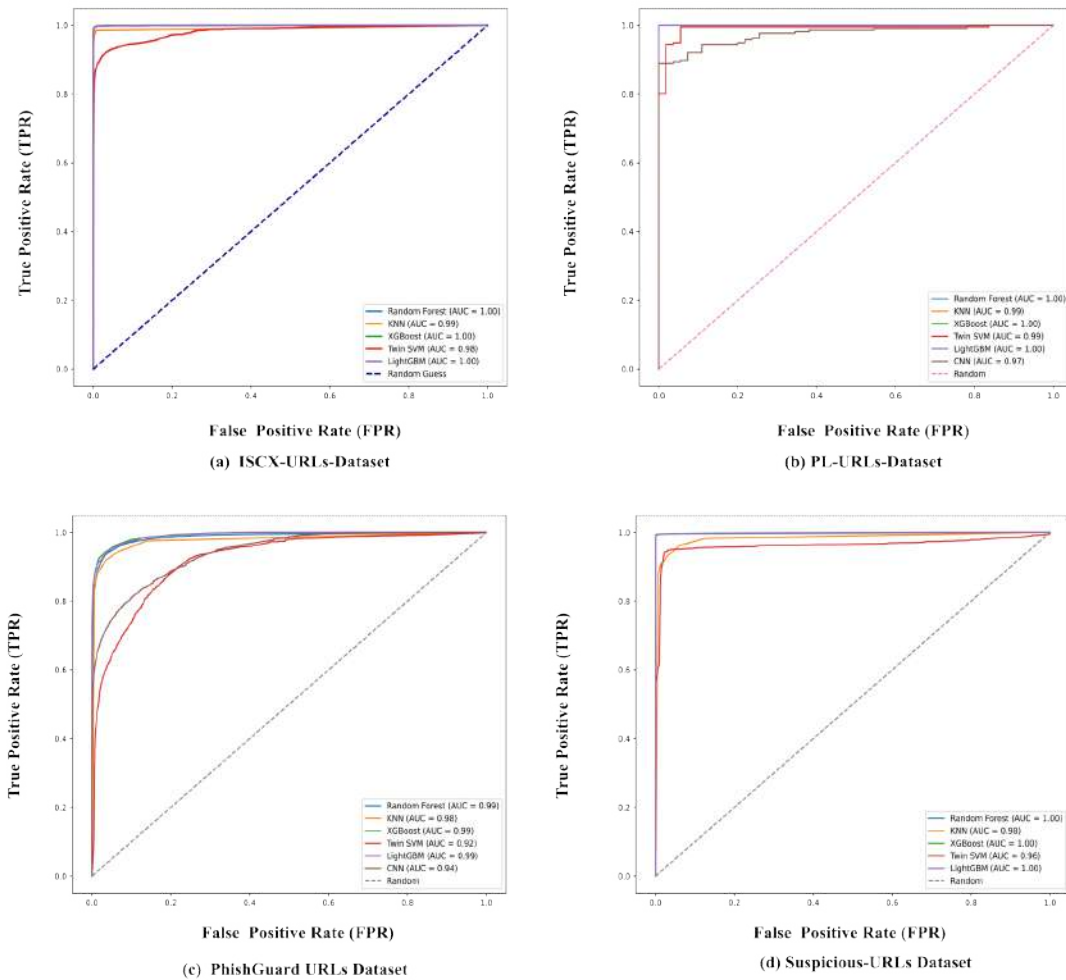


Figure 4. Confusion Matrix Analysis: ISCX-URLs Dataset (2016)

4.5.1 Comparative Benchmarking

To validate the effectiveness of the proposed phishing URL detection approach, we conducted a benchmarking analysis by comparing our results with those reported in recent state-of-the-art studies. This comparison helps contextualize the performance of our approach within the broader research landscape and highlights its advantages in terms of accuracy, robustness, and generalizability across multiple datasets.

The performance of our best model, XGBoost, is benchmarked against existing phishing detection frameworks reported in the literature. For instance, Dhanabal and Shantharajah [64] employed traditional machine learning techniques on the ISCX-URL dataset

and achieved a maximum accuracy of 93.79%, while our XGBoost model achieved 99.65% on the same dataset, reflecting a significant improvement. Similarly, the method proposed by Basnet et al. [65] using decision trees reported an F-measure of 92.45%, which is outperformed by our model with a peak F-measure of 99.12% on ISCX-URLs.

In recent work, Saiedi et al. [66] integrated deep learning techniques and achieved an AUC of 95.20% on hybrid datasets. In contrast, our model consistently surpassed this benchmark, achieving AUC scores of up to 99.60% across multiple datasets including ISCX-URLs, PL-URLs, and Suspicious-URLs. Furthermore, the lightweight phishing detection model presented by Hendaoui et al. [67] reported a TPR of 97.12% and

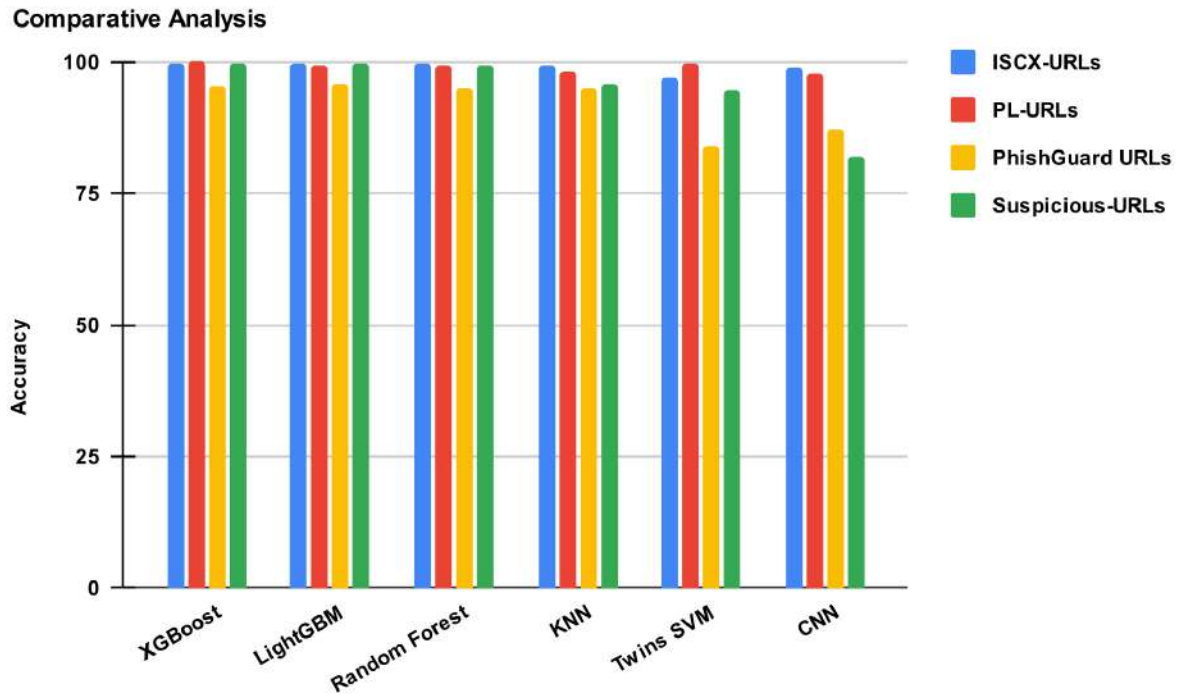


Figure 5. Comparative analysis of the experimental results of the proposed approach.

an FPR of 2.34% on the PhishTank dataset. In contrast, our model achieved a TPR of 99.92% and FPR of just 0.74% on the more diverse Suspicious-URLs dataset.

Table 7 summarizes the comparison of our model's performance with previously published results. The consistent superiority of our XGBoost model across multiple evaluation metrics and datasets demonstrates its practical utility and reinforces the reliability of the proposed methodology.

The results confirm that our proposed model significantly outperforms existing approaches across various metrics, offering a more accurate and explainable phishing detection solution. The integration of LIME further strengthens model transparency, which is essential for deployment in sensitive cybersecurity domains.

4.6 LIME analysis

This research study analyzed individual predictions using LIME to uncover the localized reasoning behind the model's classifications. Specifically, LIME was used to clarify the prediction of instances as either

Legitimate or Phishing. As illustrated in Fig. 6 LIME provides detailed explanations for model predictions on specific instances under investigation. Fig. 6 (a) presents the LIME explanation for the fifth instance predicted as legitimate by the XGBoost Algorithm. It highlights features such as "Domain Length", "No. of delimiters in Domain", and "No. of Dots in URL" are significant contributors to this classification. This instance-specific analysis allows for understanding and validating the importance of these features with the model's local decision-making patterns. Fig. 6 (b) visualizes the LIME explanation for the second instance predicted as legitimate by the XGBoost Algorithm. In this case, features like "Domain Entropy", "Domain Length", and "URL Path Length" emerged as key factors influencing the model's classification, further demonstrating the interpretability of the predictions.

Fig. 7 (a) illustrates the LIME explanation for the fourth instance predicted as legitimate by the XGBoost Algorithm. We identified features such as "Presence of



Figure 6. LIME-generated explanations for XGBoost predictions on ISCX-URLs and PL-URLs datasets.

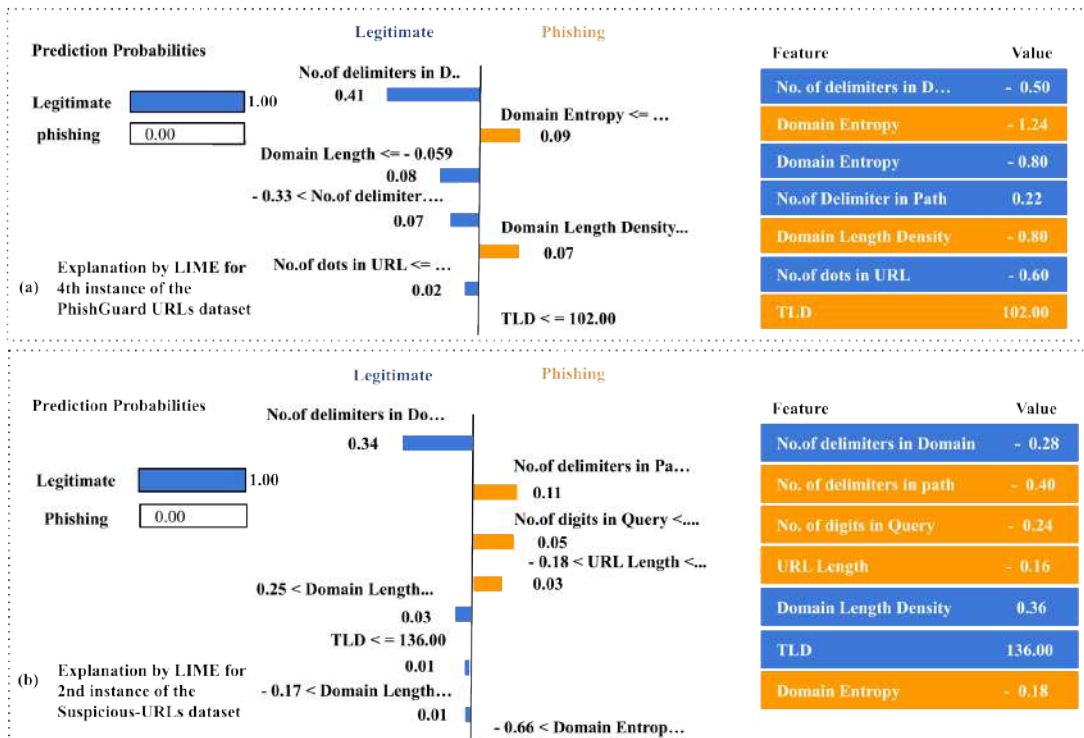


Figure 7. LIME-generated local explanations for XGBoost predictions on PhishGaud and Suspicious-URLs datasets

Table 7. Benchmark comparison with existing phishing detection approaches.

Study	Technique	Dataset	Accuracy (%)	AUC (%)	FPR (%)
[64]	Traditional ML (SVM, DT)	ISCX-URL	93.79	92.10	4.7
[65]	Decision Tree + Feature Engineering	Self-collected	92.63	91.00	5.1
[66]	Deep Learning	Hybrid dataset	95.20	95.20	3.85
[67]	Lightweight ML (XAI-integrated)	PhishTank	97.80	97.12	2.34
This Study (XGBoost)	XGBoost + LIME	ISCX-URLs	99.65	99.60	0.37
This Study (XGBoost)	XGBoost + LIME	Suspicious-URLs	99.59	99.59	0.74

Special Characters in Domain", "Domain Length", and "SSL Certificate Validity" as critical to the prediction, showcasing the localized feature relevance for this dataset. Fig. 7 (b) depicts the LIME explanation for the second instance predicted as legitimate by the XGBoost Algorithm. Here, features like "Number of Suspicious Keywords in URL", "No. of Delimiters in Domain", and "Presence of IP Address in URL" were influential, validating the model's decision-making rationale for this dataset. Across all instances and datasets, we observed a consistent reliance on certain key features, such as "No. of delimiters in domain" and "Domain Length," in phishing detection. This consistency underscores the robustness of the XGBoost Algorithm in utilizing critical features to make accurate predictions.

This comprehensive analysis using LIME provides transparency into the individual predictions made by the XGBoost Algorithm and the importance of key features across different datasets. These findings demonstrate the value of interpretability techniques in ML, offering deeper insight into model behavior and enhancing the trustworthiness of automated decision-making systems. In conclusion, LIME has proven to be an effective tool for explaining the reasoning behind model predictions. It underscores the importance of consistent feature relevance in phishing detection and enhances the interpretability of models in complex classification tasks.

4.7 Discussion

Despite the emerging nature of ML in phishing detection, its application to analyze URL patterns is highly effective. ML can provide CyberSecurity

professionals with enhanced decision-making capabilities by offering predictions, classifications, and visualizations of crucial URL aspects. Unfortunately, most ML algorithms are inherently black-box, making explaining their internal decision-making processes difficult. This lack of interpretability poses a significant challenge when implementing ML-based solutions in safety-critical domains such as cybersecurity, where transparency and trustworthiness are paramount. To address this challenge, we applied six ML algorithms and utilized LIME to obtain interpretable results in the phishing attack detection process. This research concludes that LIME is an appropriate tool for visualizing and understanding the internal decision-making processes of classification models in URL data analysis. In addition, our proposed approach achieves a precision above 99% for URL data analysis, as shown in Table 5 and Table 6.

More recently, Karim et al. [22] introduced an ML-based approach for phishing detection from URLs. They proposed a hybrid LSD model that combines logistic regression, support vector machine, and decision tree (LR+SVC+DT) through soft and hard voting mechanisms, achieving an accuracy of approximately 95%. In contrast, our approach demonstrated an improved accuracy of up to 99%, incorporating LIME explanations to effectively visualize and interpret the internal decision-making processes of the ML models. In another study, Recent work by Smith et al. [47] proposed an approach to enhance classifier performance by integrating genetic algorithms with particle swarm optimization. Their study found that a multinomial Naive Bayes classifier combined with a

genetic algorithm yielded the most effective results. A comparative analysis demonstrated the superiority of this hybrid method over other ML and bio-inspired models. Building on this foundation, our study conducts performance comparisons of six ML algorithms, further enriching the analysis by employing the LIME technique to gain interpretability into the models' internal decision-making processes.

Chawla et al. [48] aimed to identify standard features of phishing websites and create a detection model using feature extraction and pre-processing on seven email datasets. Their study found that Random Forest Max Vote Classifier, Decision Tree, and Artificial Neural Network models achieved an accuracy of 97.73%. In our study, eleven features were extracted, and the overall accuracy remained at 98%, with the XGBoost algorithm showing the highest accuracy. Gupta et al. [49] emphasized the identification of essential website features such as URL structure, domain age, SSL certificate validity, page length, and the presence of JavaScript to train machine learning models for phishing detection. Their findings showed that the Random Forest classifier delivered the highest accuracy, demonstrating its robustness and reliability in detecting phishing websites using interpretable and domain-specific attributes. Kumar et al. [50] introduced a lightweight software-based approach to distinguish between legitimate and phishing websites by analyzing only the URL. Their model, built using the Random Forest classifier, achieved a 97% F1-score, 97% accuracy, and 99% recall—outperforming prior methods in both speed and efficiency by eliminating the need for additional data sources. In our study, eleven hyperlink-based features were extracted, and the XGBoost algorithm achieved the highest classification accuracy of 98%, demonstrating improved performance over conventional techniques.

More specifically, this research demonstrates the significant potential of ML in phishing detection by analyzing URL patterns. Using LIME enhances the interpretability of model decisions, providing CyberSecurity professionals with valuable insights into the classification process. Our study, supported by comparisons with existing literature, confirms that features such as

'Domain Length' and 'No. of delimiters in Domain' are critical in detecting phishing attempts, reinforcing the effectiveness of our proposed approach.

Intuitively, ML and XAI are transformative technologies in the domain of CyberSecurity, significantly advancing the ability to detect and mitigate threats. ML models, including deep learning, logistic regression, and anomaly detection algorithms, excel at processing vast and complex datasets to uncover patterns indicative of cyber threats. These include phishing, ransomware, and zero-day vulnerabilities. By automating and augmenting traditional security measures, these models enhance detection speed and accuracy, reducing the window of exposure to malicious activities. However, the inherently black-box nature of many ML models often limits their application in scenarios where interpretability and accountability are critical, such as incident response and forensic analysis.

On the contrary, XAI bridges this gap by providing methodologies to make ML models more interpretable and transparent. By explaining the reasoning behind alerts or classifications, XAI enables security analysts to validate and refine the outputs of automated systems, fostering trust and reliability. Moreover, adopting XAI aligns CyberSecurity solutions with regulatory frameworks and emphasises transparency and accountability in automated decision-making processes. By offering insights into the rationale behind automated decisions, XAI facilitates compliance and ensures the ethical use of ML technologies, reinforcing their role as indispensable tools in safeguarding digital ecosystems.

It is essential to state that the results of this study may not generalize to a larger scale due to the use of only four publicly available datasets. Therefore, further work comparing results with extended datasets from other sources would be beneficial. Despite this limitation, the outcomes achieved remain significant in CyberSecurity data analysis; the results of this research study provide valuable insights for enhancing the decision-making processes of CyberSecurity professionals. The application of ML and XAI techniques, such as LIME, provides valuable insights into the classification of phishing URLs, helping to im-

prove the decision-making processes of CyberSecurity professionals.

Future research should focus on validating the findings with extended datasets to ensure broader applicability and robustness, particularly by including diverse and multilingual datasets that reflect evolving phishing strategies. Efforts should also be directed toward developing real-time detection systems capable of high accuracy with minimal latency and adaptable for resource-constrained environments like mobile and IoT devices. Addressing advanced threats such as spear-phishing and deep-fake impersonation requires incorporating behavioural analysis and multimodal data fusion techniques. Enhancing explainability remains critical for transparency, particularly in regulatory contexts, with tools like XAI providing real-time interpretability for CyberSecurity analysts. Moreover, the scope should expand to detect phishing across various platforms, including social media and messaging apps, while ensuring resilience against adversarial attacks to maintain model integrity and trustworthiness.

5 Conclusion

The constant discovery of new technologies and methods applicable in a wide range of scenarios has accompanied the exponential growth of internet users. Numerous websites with login pages are created daily, often attempting to obtain users' information illicitly. The sheer number of websites makes it increasingly difficult to discern safe sites. Users who interact with these emails or links are redirected to fraudulent websites containing malicious links, inadvertently providing their private information to attackers. This research focused on enhancing the detection of phishing attacks through feature engineering, ML algorithms, and XAI. According to experimental results, XGBoost outperformed other ML algorithms in phishing attack detection. The study integrates XAI to mitigate the black-box nature of ML algorithms, aiming to provide the research community and CyberSecurity professionals with deeper insights into the underlying patterns of malicious phishing URLs. This understanding is crucial for improving security

measures and developing robust defences against phishing threats. The main contributions of this research include identifying the most effective feature engineering techniques for phishing attack detection, performing comparative performance analysis of various ML algorithms, and utilizing XAI to elucidate the inner workings of ML algorithms in identifying essential features in phishing attack detection.

In conclusion, this study demonstrates the effectiveness of integrating ML and explainable AI techniques for phishing detection, achieving high accuracy and interpretability across diverse datasets. The proposed methods address the inherent challenges of black-box models by leveraging XAI to enhance transparency and decision-making. However, future research should focus on validating these findings with extended and diverse datasets to ensure robustness and broader applicability. Emphasis should also be placed on developing real-time detection systems adaptable to resource-constrained environments and addressing advanced threats like spear-phishing and deep-fake impersonation through behavioural analysis and multimodal data fusion. Expanding detection capabilities to platforms such as social media and ensuring resilience against adversarial attacks will further strengthen CyberSecurity defences in the evolving digital landscape.

Author Contributions

Muhammad Nauman: Conceptualization, Methodology, Software **Hafiz Muhammad Usman Akhtar:** Main Contribution: Conceptualized the study and led the research direction; Data Curation: Collected, cleaned, and organized datasets; Writing – Original Draft Preparation: Authored the initial manuscript draft **Mustafa Hameed:** Visualization, Investigation. **Nadeem Akhtar:** Supervision.: **Muhammad Zee-shan Tareen:** Software, Validation. **Sidra Hameed:** Writing- Reviewing and Editing.

Compliance with Ethical Standards

It is declare that all authors don't have any conflict of interest. It is also declare that this article does not contain any studies with human participants or animals performed by any of the authors. Furthermore,

informed consent was obtained from all individual participants included in the study.

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